

Graph Mining Course Project Proposal

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Course: Graph Mining 4041

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Project Title

An Active Diffusion Neural Network for Graphs (ADGNN)

Abstract

Graph-structured data such as citation and social networks are central in graph mining, and semi-supervised node classification is a core task. Many Graph Neural Networks (GNNs) can be viewed as passive diffusion processes that repeatedly smooth node features, which often leads to over-smoothing in deeper models. The recent Active Diffusion Neural Network for Graphs (ADGNN) treats diffusion as an active process by adding several external source terms to the diffusion equation. In this project We will implement a simplified ADGNN layer for node classification, compare it with a two-layer GCN and a diffusion-based baseline on citation graphs, and study how active diffusion affects classification accuracy and over-smoothing.

Problem and Motivation

Node classification on graphs arises in applications such as paper recommendation, community analysis and fraud detection. Standard message-passing GNNs, including Graph Convolutional Networks (GCNs), mix information between neighbours in a way that resembles heat diffusion. On homophilic graphs this can help propagate label information, but deeper or more strongly diffusive models tend to suffer from over-smoothing, where node embeddings become nearly identical and labels are hard to separate. Diffusion-based variants such as personalised-PageRank GCNs partly address this by changing the propagation kernel, yet they remain passive: information only flows along edges and no new signal is injected. The ADGNN framework is interesting for this course because it explicitly augments diffusion with multiple external sources. Studying a simplified ADGNN gives a concrete way to connect graph Laplacians, diffusion dynamics and semi-supervised classification within a manageable undergraduate-level project.

Objectives

- Implement a two-layer GCN and one diffusion-based GNN baseline for semi-supervised node classification.
- Design and implement a simplified ADGNN layer with multiple external source terms.
- Compare ADGNN with baselines on citation graphs in terms of accuracy and over-smoothing behaviour.

Related Work

GCNs by Kipf and Welling are the standard baseline for semi-supervised node classification and can be interpreted as a first-order approximation to spectral graph convolution. Diffusion-based methods such as Diffusion Improves Graph Learning (GDC) pre-compute propagation matrices using personalised PageRank or heat kernels, while GRAND and related neural diffusion models simulate feature diffusion with an additional source term that preserves original features. ADGNN extends this line of work by introducing three distinct source terms—ego, boundary and anomaly components—combined with a normalised adjacency term in a closed-form diffusion solution. This project will empirically compare a streamlined ADGNN with GCN and one diffusion-based baseline on benchmark graphs.

Proposed Methodology

Implementation will be in Python using PyTorch and PyTorch Geometric. We will first implement a standard two-layer GCN and verify that it reaches reasonable accuracy on a citation dataset. As a diffusion-based baseline We will add a GCN variant with a fixed personalised-PageRank or heat-kernel propagation matrix. Building on these baselines, We will design a simplified ADGNN layer that combines four components at each step: an ego term from a linear transform of the original node features, a boundary-aware term, an anomaly-sensitive Laplacian term and a normalised adjacency diffusion term. Each component will be weighted by a learnable scalar, and the layer will be applied a small fixed number of times to approximate the infinite-diffusion solution suggested in the paper. We will then integrate this layer into a shallow network and train it end-to-end on the chosen datasets.

Dataset(s)

We plan to use the standard citation benchmarks Cora, CiteSeer and PubMed, which contain a few thousand nodes, sparse bag-of-words features and between 3 and 7 classes. These graphs are directly available in PyTorch Geometric with predefined train/validation/test splits, so only minimal preprocessing such as feature normalisation or removal of isolated nodes should be necessary.

Techniques and Algorithms

- Two-layer GCN as a passive diffusion baseline for node classification.
- One diffusion-based baseline using a precomputed personalised-PageRank or heat-kernel propagation matrix (GDC-style).

- Simplified ADGNN layer combining ego, boundary, anomaly and adjacency diffusion terms with learnable scalar weights.
- Python, PyTorch, PyTorch Geometric and optionally NetworkX for implementation and graph inspection.

Evaluation Plan

We will evaluate all models on semi-supervised node classification using test accuracy (and optionally macro-F1) as the main metric, averaging results over several random seeds. To analyse over-smoothing, We will measure the average cosine similarity between embeddings of nodes with different labels as the effective number of diffusion steps increases, and optionally visualise embeddings with a two-dimensional projection. GCN, the diffusion-based baseline and the ADGNN variant will be compared under shared training settings.

Challenges and Resources

The main challenge is implementing the active diffusion update rule in a stable and efficient way while keeping the architecture simple enough for the course timeline. We will mitigate this by starting from well-tested GCN examples in PyTorch Geometric and validating each additional term on small toy graphs. A second challenge is managing experiment time; We plan to restrict the number of datasets and runs and use early stopping on a single GPU-equipped machine.

References

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